Cleaning the official data

Methodology

1. The main database was received as a response to an information request to the INM filed in the Plataforma Nacional de Transparencia. It covers the period from 2017-2023 and the first half of 2024.
2. The following information was received in eight tables, organized by year. They had the following categories:
    - Oficina de Representación del INM - Regional INM office
    - Lugar de origen - Place of origin
    - Lugar de destino - Destination
    - Total de migrantes en traslado - Number of immigrants aboard
    - Nacionalidad - Nationality
    - Hombres - Men
    - Mujeres - Women
    - Niñas - Girls
    - Niños - Boys
3. To these initial columns, we added the columns id_fila_og, which contains a unique ID number for each original row, and id_viaje, which contains a unique ID for each trip, which can span multiple original rows due to how the data was formatted.
4. The automatic cleaning of the tables was not straightforward; certain cells were combined to created overarching tags for the data, while others were combined to permit a more precise breakdown of the categories requested. The meaning of how cells were combined and divided also varied greatly from year to year and between participating regional INM offices. As such, the first step we took was to identify the rules each regional office followed to compile its data through a close reading of the database.
5. First, the specific rules were identified, along with the data ranges to which they corresponded. Next, all of the specific rules were examined to see if some could be combined and resolved through the same algorithm, in order to create a more efficient process. Finally, the column “regla” was created and each row was designated with the rule it would follow to be concatenated into the master table with all 8 years of data. Originally, 11 specific rules were identified, but through this process we could combine them into 4 rules and 2 special cases.
6. While the complete documentation of the original tables’ rules is linked in this footnote,[1] here we will mention a few of the interpretative decisions made in order to concatenate the tables, which became necessary when there were what seemed to be typos or inconsistencies in the original data. For example, the data of the Coahuila regional office in 2022 seem to be copy-pasted one column to the left of where they should have been. This error was interpreted and corrected as a typo in order to be able to conserve the majority of the data.
7. In the 2023 table, the column “Lugar de origen” was named “Oficina de Representación del INM”, creating a duplicate column in what seemed to be a typo. The places of origin were still largely broken down into localities in this table, but others were more vague, only repeating the name of the regional office; for example, Nuevo León and Estado de México. For such entries, the real-world addresses of each regional office were located in order to be able to conserve the data and visualize the transfers by municipality; for example, Nuevo León became Guadalupe, Nuevo León[2] and Estado de México became Toluca, Estado de México.[3]

8. In the 2024 table, various Michoacán entries only show Mexicans being transferred, which may represent US deportations of Mexican nationals to Michoacán. These rows weren't eliminated from our analysis, since they appear to form part of the infrastructure of return and deportation routes, but they are a noteworthy case within the trend.
9. After applying the rules, rows were eliminated if they still had blank values for lugar_origen, lugar_destino, total_migrantes_traslado, and/or nacionalidad. Rows for which lugar_origen, lugar_destino or nacionalidad had a number instead of a string (for example the rows with fila_id_og IDs 2778 to 2780) were also eliminated. And rows were checked and corrected if total_migrantes_traslados was a string instead of a number. If a row lacked all demographic data, it was eliminated; if just a few of the numbers within the sex and age breakdown were missing, for example in the case of Puebla in 2024, the row was conserved and the blank data was filled with 0's.
10. There are rows for which the sum of the demographic categories registered doesn't coincide with total_migrantes_traslado, but we decided to assume that the total was correct and that the errors present were more likely in the demographic breakdown. The decision to maintain rows with this irregular data as they came was made in recognition that for the analysis at hand, the demographic data received was broken down in excess of what was strictly required; we only needed to verify the presence or absence (binary value) and the total number (sum) of boys and girls on each transfer. If there were errors in the number of children registered, comparing it would total_migrantes_traslado would do nothing to resolve it, so we stuck with the information we were provided.
11. Once the full table was compiled, munipality and state names were standardized using Python. The majority of these names could be assigned exactly, but origins and destinations registered to "Ciudad de México" or "CDMX" were divided between the municipalities of Cuauhtémoc and Iztapalapa, which is an estimation.

Limitations

The variations in how different regional offices organized their data, as well as the discrepancies between the totals of the demographic breakdown versus the column total_migrantes_traslado, together suggest a highly manual and discretional documentation culture in the INM, producing archives with low levels of internal standarization. We don't have a way to find out the methodology that guided the creation of these records, but we tried to compensate for their inconsistencies by taking care to document our own process of concatenation and attempting to recuperate the greatest possible instance of relevant data. Even so, we recognize that our data represent an estimation of the historical reality rather than an exact portrait.

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1. Frankel, 2026. "Reglas tablas". https://docs.google.com/spreadsheets/d/17OCfRHTOGVZ7WL_YxtLJe7dqQxjmPX-1wGMM3uXzsgY/edit?gid=1896445730#gid=1896445730
2. "Nuevo León". Gobierno de México. https://www.inm.gob.mx/gobmx/word/index.php/nuevo-leon/
3. "Estado de México". Gobierno de México. https://www.inm.gob.mx/gobmx/word/index.php/estado-de-mexico/

Creating the map of routes

Purpose and final product:

- We wanted to create a map with nodes and lines representing the key points through which transit migration in Mexico has historically tended to pass
- The routes mapped should reflect as closely as possible the real routes of the freight train and interstate highway routes that represent the most efficient ways that people on the move could transit Mexico from south to north; in other words, they should represent paths of least resistance
- The route analysis should allow us to consistently analyze:
    - the direction, in relation of the map of routes from north to south, of each INM transfer
    - if the destination of each transfer was along one of the main historical transit migration routes, on an alternate route, or outside of any documented historical route; in other words, the correspondence of each INM transfer with the map of historical routes

Methodology:

1. Nodes and their connections were identified with reference to:
    - our own fieldwork
    - the fastest highway routes between key points as generated by the algorithms of Google Maps
    - the Map of Routes, Assistance and Risks Associated with Irregular Migration [RARAMIR][1] maintained by the Mexican government secretary's Immigration Policy Unit
    - a variety of historical maps of transit migration routes through Mexico, for example the map in a report by Amnesty International Mexico (2010),[2] the map in a Red de Fondos Ambientales de Latinoamérica y el Caribe report (2024),[3] and the map included in the book *Walking Together: Central Americans and Transit Migration through Mexico.*[4]
2. The nodes were loaded as points in QGIS with reference to the base map provided by Open Street Maps.
3. The official open route databases were downloaded from the National Railway Network [RFN] (2024) and the National Network of Roads [RNC] (2025), then uploaded to QGIS.
4. The railway routes were filtered to only include freight routes currently in operation. The road routes were filtered to only include federal and state highways; later, these filtered routes were combnied and cleaned in QGIS and GRASS to form a continuous network, creating the base network vías_principales.
5. To each key point we identified, we added a peripheral ring which looped its center point at a distance of 5-10 kilometers, depending on the city's size, so that the routing algorithm could select the most efficient entry and exit points to the city as a whole, independently of its center point. These peripheral rings were incorporated to vías_principales.
6. The routes were generated by segments between each of the key points identified using QGIS's Network Analysis function. In the few cases where it wasn't possible to generate a route using just vías_principales, we added back the smaller roads omitted in step 4.
7. Each route segment was checked visually to ensure its basic correspondence with vías_principales:
    - The majority of the routes generated followed vías_principales exactly.

- Almost all of the deviations observed followed either smaller roads or the auto-generated peripheral rings for very short distances.
- Of the very few remaining deviations that didn't follow either a smaller road or a peripheral ring, and were more likely the result of snaps or the cleaning of intersections, none was so drastic as to cross municipal lines.

8. The route segments generated were combined in four long routes from the southern border to central Mexico [Pacífico, Chiapas, Tabasco, Caribe], and five long routes from central Mexico to the northern border with the US [Baja, Sonora, Chihuahua, Texas, Golfo]. The southern routes are differentiated by their origins, and the northern routes, by their destinations. However, in central Mexico, many of the routes overlap.
9. Any of the southern routes can connect to any of the northern routes, generating 20 possible routes from one border to the other. However, these possible route combinations can be organized into two categories. The Pacific route can be assumed to be the main route leaving the southern border, since its origins are the most highly transited border crossings. For any northern route – Baja, Sonora, Chihuahua, Texas, or Golfo – we can assume the Pacific route as its probable origin. The alternative southern routes, Chiapas, Tabasco and Caribe, generally have a secondary weight in the analysis, with the exception that for the northern Golfo route, the southern Tabasco route can also be considered a probable origin, since this combination is the literal shortest route from south to north. The northern routes were all weighted equally, since immigrants' choice of crossing point to the US usually responds to their personal assessment of the security situation, where their family in the US already lives, and highly changeable online rumors about US border security. As such:
    - The main routes are: pacifico_baja, pacifico_sonora, pacifico_chihuahua, pacifico_texas, pacifico_golfo and tabasco_golfo;
    - The alternative routes are: chiapas_baja, chiapas_sonora, chiapas_chihuahua, chiapas_texas, chiapas_golfo, tabasco_baja, tabasco_sonora, tabasco_chihuahua, tabasco_texas, caribe_baja, caribe_sonora, caribe_chihuahua, caribe_texas, caribe_golfo.
10. The southern and northern routes connect in a few cities in central Mexico; the case of Puebla was the most complex, since it was originally classified as part of both the northern and the southern routes. In order to avoid confusions, it was later reclassified as part of the northern routes.

Limitations:

In the RNC and RFC databases, there are no indicators of direction in the routes that became the basis for vías_principales. However, we assume that since the routes selected were almost entirely main thoroughfares, there were likely few one-way routes among them; we also assume that any one-way routes were likely doubled by another one-way route in the other direction that didn't present many differences in the order of the municipalities it crossed.

The route map and designation of main versus alternative routes were designed to show paths of least resistance while consciously omitting an important factor: how people on the move tend to change their minds. We only mapped the connections between nodes in the order that they would be passed through by a person who, starting from their point of origin on the southern border, already had firmly in mind the point on the northern border to which they hoped to arrive, and was attempting to get there as quickly as possible.

1. Unidad de Política Migratoria, 2025. “Mapa de Rutas, Asistencia, y Riesgos Asociados a la Movilidad Irregular por México. Gobierno de México. https://politicamigratoria.segob.gob.mx/es/PoliticaMigratoria/raramir
2. Amnistía Internacional, 2010. “Víctimas Invisibles”. https://www.amnesty.org/es/wp-content/uploads/sites/4/2021/06/amr410142010es.pdf
3. Morales, 2024. “Las rutas migratorias en México”. Twitter. https://x.com/moaimx/status/1829950565947977832
4. Díaz de León, 2023. “Walking Together: Central Americans and Transit Migration through Mexico”. University of Arizona Press.

Analysis of transfers' direction, correspondence with historical routes and detention infrastructure, and regional diversity

Purpose:

- Assign the direction of each transfer with reference to the map of historical routes
- Determine the level of correspondence of each transfer with the historical route, with reference to the map generated
- Determine the regional diversity of people present on each transfer
- Indicate if there was immigrant detention infrastructure present in the origin and destination points of each transfer, using the information compiled by the CNDH in 2024[1]

Data dictionary:

For datos_menores_regiones (includes deportations):

Column name	Description
id_fila_og	Unique ID for each row that is consistent from its first appearance in the original data through every transmutation and final product
year	Year of transfer
oficina_regional	Contributing INM regional office
lugar_origen	The municipality and state of origin
lugar_destino	The destination municipality and state
total_migrantes_traslado	The number of people aboard
nacionalidades_orig	The nationalities of those aboard in their original phrasing
hombres	Number of men aboard
mujeres	Number of women aboard
niñas	Number of girls aboard
niños	Number of boys aboard
regla	The rule guiding how how this transfer was incorporated into the concatenation of the eight original tables[2]

id_viaje	Unique ID number assigned to each transfer on the basis of rows which had values for total_migrantes_traslado in the original tables; maintained from its assignation through every transmutation and final product
origen_limpio_mun	Origin municipality name
origen_limpio_ent	Origin state name
destino_limpio_mun	Destination municipality name
destino_limpio_ent	Destination state name
destino_cve	Unique ID assigned by Mexico's National System of Statistical and Geographical Information [INEGI] for the destination municipality
origen_cve	Unique ID assigned by Mexico's National System of Statistical and Geographical Information [INEGI] for the origin municipality
geom_destino	Geographical and geometric data
geom_origen	Geographical and geometric data
lon_destino	Longitude of the destination
lat_destino	Latitude of the destination
lon_origen	Longitude of the origin
lat_origen	Latitude of the origin
sentido	• **ava** if the transfer advanced those aboard along their routes • **ret** if it returned them • **lat** if it moved them laterally between parallel sites • **int** if it was within a single municipality • **dep** if it was a deportation
menores_num	Sum of the columns niñas + niños
menores_bin	1 if menores_num > 0; 0 if menores_num = 0
norteamerica	1 if in nacionalidades_orig there were North Americans; 0 if not
centroamerica	1 if in nacionalidades_orig there were Central Americans; 0 if not
sudamerica	1 if in nacionalidades_orig there were South Americans; 0 if not
caribe	1 if in nacionalidades_orig there were Caribbeans; 0 if not
extracontinentales	1 if in nacionalidades_orig there were people from outside the continent; 0 if not
nan_región	1 if in nacionalidades_orig there were people from unidentified regions; 0 if not
diversity	Sum of columns norteamerica + centroamerica + sudamerica + caribe + extracontinentales
diversity_cat	"Una región" if diversity = 1, "Dos o más regiones" if diversity > 1

Additional columns in datos_visualizables (just internal transfers):

rutas_posibles_o	List of routes on which the origin municipality is included
rutas_posibles_d	List of routes on which the destination municipality is included

correspondencia	• **ruta_cont_1** if both origin and destination municipalities are along a main route, or within the same route • **ruta_cont_2** if origin and destination municipalities are along any continuous alternative route • **rutas_diferentes** if origin and destination municipalities are on parallel routes • **d_fuera** if the destination isn't on any route • **o_fuera** if the origin isn't on any route • **ambos_fuera** if neither origin, nor destination are on a route
em_o	1 if the origin municipality has detention infrastructure; 0 if not
em_d	1 if the destination municipality has detention infrastructure; 0 if not
ems	• If em_o + em_d = 0, **no_em** • If = 1 y em_o = 1, **em_o** • If = 1 y em_d = 1, **em_d** • If = 2, **ambos**

Methodology

To determine direction

The direction of each transfer route was evaluated in reference to the previously generated map of historical routes. Its designation was manually registered, then assigned to all instances of the given transfer route, and checked visually during different phases of the visualization process.

To determine correspondence with historical routes

1. All routes were given buffers of 5 km, so that the final analysis could take nearby rest stops into account.
2. The municipalities crossed by each buffered route were extracted from the historical route map using the Select function in QGIS. In Central Mexico, the routes could coincide, meaning that a single municipality might be on multiple routes.
3. The extracted municipalities were added to a CSV and compared in Python with the origin and destination points documented for each INM transfer. The columns rutas_posibles_o and rutas_posibles_d were created to list the routes which crossed each origin and destination point.
4. Using the criteria described in the data dictionary, the column "correspondencia" was created.
5. For the final analysis of correspondence which appears in this report:
    - Transfers with the value "ruta_cont_1" were categorized as ending "Along a main route"
    - Transfers with the value "ruta_cont_2" or "rutas diferentes" were categorized as ending "Along an alternate route"
    - Transfers with the value "d_fuera" or "ambos_fuera" were categorized as ending "Outside established routes"
    - Transfers with the value "o_fuera" were categorized as ending "Along a main route" if their destination was along a main route, and categorized as ending "Along an alternate route" if their destination was only along an alternate route

Para describe transfers' correspondence with maps of detention infrastructure

1. The reference table "EMs" was created using CNDH data. It shows where the INM has reported detention infrastructure, including each center's municipality, state, CVEGEO, and estimated year inaugurated. Some detention centers didn't contain information about their inauguration date; these were left as 0s.
2. For each transfer, the municipalities of origin and destination were compared with EMs to check if they had any kind of detention infrastructure documented as operating there during the year the transfer took place. Depending on the result, they were assigned values for the columns em_o, em_d, and ems as documented in the data dictionary.

Limitations:

Los datos el INM son inexactos; una colega que realizó una solicitud similar a la nuestra sobre los traslados de personas migrantes recibió una base de datos también extensa, pero con datos diferentes; y el CNDH documenta que la información proporcionada por el INM respecto a las estaciones migratorias difirió de la misma información que la Comisión solicitó del INM en 2019. Parecería que el mismo INM no cuenta con registros consistentes y facilmente asequibles respecto a las operaciones de todos sus sitios de detención migratoria, lo que resalta que otra vez, este análisis describe y estima la realidad, pero no podemos afirmar que lo retrate en cada detalle.

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1. CNDH, 2024. "Informe Especial sobre las Condiciones de las Estancias y Estaciones Migratorias: Hacia un Nuevo Modelo para la Atención de la Migración Irregular". CNDH. https://www.cndh.org.mx/sites/default/files/documentos/2024-02/INFORME%20ESPECIAL%20ESTANCIAS%20MIGRATORIAS.pdf
2. Frankel, 2026. "Reglas tablas". https://docs.google.com/spreadsheets/d/17OCfRHTOGVZ7WL_YxtLJe7dqQxjmPX-1wGMM3uXzsgY/edit?gid=1896445730#gid=1896445730